Politecnico di Milano

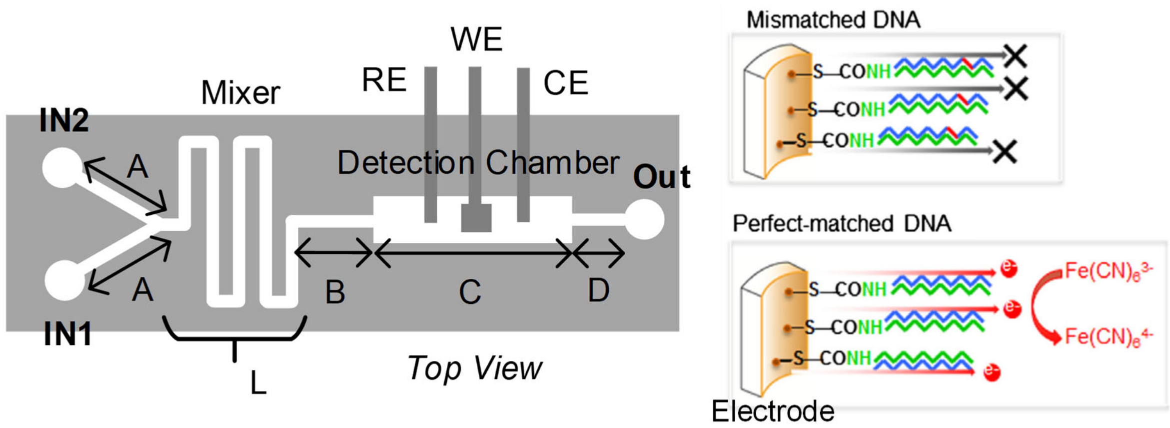
Biochip A.Y. 2020/2021 - M. Carminati

Exam of 9.02.2022

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

Exercise 1

Consider the following microfluidic device for the detection of DNA strands. DNA fragments (IN1) are mixed with a redox molecule (FeCN in IN2) and electrochemically detected in the chamber C:



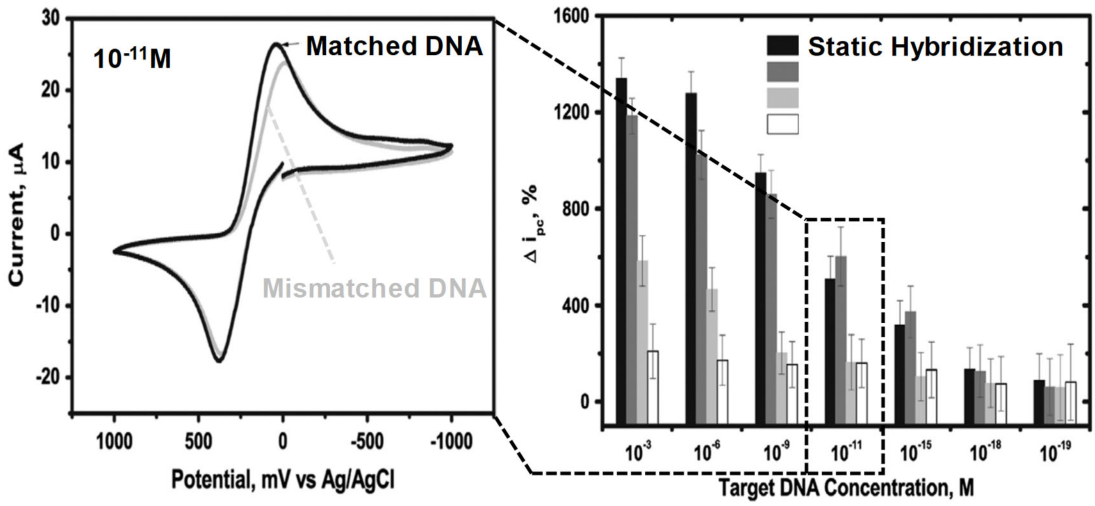
Data:

$h = 20\ \mu m$
$w_A = w_B = w_L$
$w_D = 25\ \mu m$
$w_C = 100\ \mu m$
$A = 1\ mm$
$L = 10\ mm$
$B = 2\ mm$
$C = 3\ mm$
$D = 0.5\ mm$
$1 atm = 101\ kPa$
$\eta_{water} = 10^{-3}\ Pa{\cdot}s$

a) Draw the electrical fluidic equivalent of the system and find the value of its parameters.
b) Assuming P_{IN1} = 2 atm, P_{IN2} = 1.8 atm, P_{OUT} = 1 atm find the fluid speed in all sections of the device.
c) Write the complementary DNA strand to this sequence: ATTCGAG. How many base-pairs can be detected if the max. length of the probe strand is 6 nm?
d) Where should a heater be placed in order to control hybridization? What temperature range should be generated by the heater?
e) Propose a way to move DNA in the device alternative to the use of a pump.
f) List the steps of a fabrication procedure of the device considering electrodes on the bottom of the channel.

Exercise 2

Now focus on the electrochemical detection of DNA: when a matched DNA strand binds with the probe the redox current between electrode and $Fe(CN)_6$ molecules increases as shown in the picture (consider static hybridization):



Data:

$V_{DD} = +5\ V$
$V_{SS} = -5\ V$

$\varepsilon_0 = 8.86{\cdot}10^{-12}\ F/m$
$\varepsilon_{r\_water} = 80$
$\delta_{PBS} = 1.5\ S/m$
$C_{DL} = 0.1\ pF/\mu m^2$

$e_n^2 = \left(4nV/\sqrt{Hz}\right)^2$

a) Draw the scheme of a **potentiostat** connected to the 3 electrodes and define the properties of the applied voltage stimulation waveform.
b) Assuming an area of the working electrode (WE) of 400 μm^2, PBS as supporting electrolyte and neglecting the CE impedance, find the small-signal equivalent impedance of the interface, including the redox term (quantify terms when possible).
c) Size the **TIA** of the potentiostat in order to cover a DNA concentration range from 10^{-9} down to 10^{-18} Molar. What is the maximum **scan rate** allowed by this circuit?
d) Considering the opamp equivalent voltage noise, resistors contributions and a total capacitance of 10 pF plot the input referred **noise spectral density**.
e) If cyclic voltammetry is to be replaced by **amperometry** for detection, what potential should be set?